# Practical-1

## SUMMARY-

In this practical, I learned how different sorting algorithms arrange data in ascending order using different approaches.

**Bubble Sort** repeatedly compares and swaps adjacent elements,

**Selection Sort** selects the smallest element and places it in the correct position,

**Insertion Sort** inserts each element into its proper place in the sorted part of the array,

**Quick Sort** uses a pivot to divide the array into smaller parts for faster sorting, and

**Merge Sort** follows the divide-and-conquer approach by splitting and merging arrays.

By implementing these algorithms, I understood their working process, logic, and differences in performance. This practical helped me improve my programming skills and gave me a better understanding of sorting techniques.

## Conclusion-

From this practical, I concluded that every sorting algorithm has its own advantages and limitations. Bubble, Selection, and Insertion Sort are simple and easy to understand but are less efficient for large datasets. Quick Sort and Merge Sort are much faster and more suitable for handling large amounts of data because of their efficient divide-and-conquer approach. Overall, this practical improved my understanding of sorting algorithms and showed me the importance of choosing the right algorithm based on the size and requirements of the data.